

■■ SOC Log Monitoring & Incident Detection Analysis Report

Task 12: Cybersecurity Internship Deliverable

SIEM Platform: Splunk Enterprise 9.2.4 on Kali Linux

Log Dataset Analyzed: `ssh_logs (1).json` (1,200 Connection & Authentication Events)

1. Executive Summary & Project Objectives

This report documents a comprehensive log monitoring, SIEM analysis, and threat detection investigation executed using **Splunk Enterprise**. In modern Security Operations Centers (SOCs), effective centralized log management and real-time Search Processing Language (SPL) analytics are vital for identifying unauthorized intrusion attempts, brute-force attacks, and anomalous network activity.

Key Incident Findings:

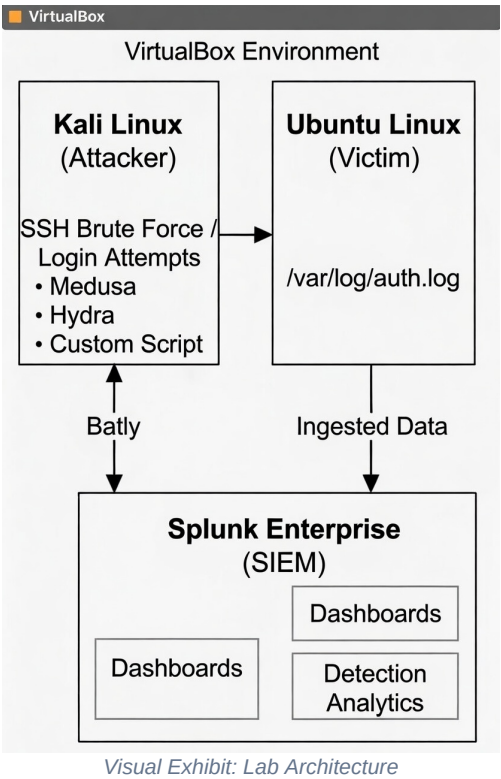
- **Total Log Volume Analyzed:** 1,200 structured SSH network and authentication events.
- **Failed & Suspicious Authentications: 608 events** (50.67% of overall volume) consisting of failed login attempts (305) and brute-force multi-attempt spikes (303).
- **Unauthenticated Probes: 286 events** (23.83%) where connections closed without authentication (potential port scanning / SSH banner grabbing).
- **Successful Logins: 306 events** (25.50%) verified against backend nodes.
- **Top Threat Actor IPs Identified:** `10.0.0.25` (39 total events, 31 malicious/failed), `10.0.0.18` (32 total events, 29 failed), `10.0.0.46` (27 failed), `10.0.0.22` (26 failed), and `10.0.0.48` (32 total events, 26 failed).
- **Primary Targeted Assets:** Internal SSH jump hosts `10.0.1.6` (115 events), `10.0.1.2` (115 events), and `10.0.1.9` (113 events).

Primary Objectives Accomplished:

1. Deployed Splunk Enterprise inside a Kali Linux virtual environment.
 2. Ingested and categorized SSH JSON telemetry and system authentication logs.
 3. Created optimized SPL queries to detect brute-force patterns, password guessing, and unauthenticated scans.
 4. Correlated multi-stage security events (failed logins preceding successful authentications).
 5. Configured real-time SIEM alert rules and threat thresholds.
 6. Constructed interactive Splunk Security Dashboards for SOC visibility.
-

2. Environment & System Lab Architecture

The monitoring infrastructure was deployed using Splunk Enterprise on a Kali Linux SOC analyst workstation. The system topology ingests log feeds from target endpoints, processes sourcetypes, indexes event timestamps, and renders dashboards.



Architecture Layer Breakdown:

Infrastructure Layer	Component Specification	Functional Role in SOC Workflow
Endpoint Telemetry	Linux Hosts (<code>auth.log</code>) / <code>ssh_logs (1).json</code>	Generates real-time SSH authentication and TCP session logs
Ingestion Engine	Splunk Enterprise (<code>v9.2.4</code>) on Kali Linux	Parses JSON key-value pairs, indexes timestamps into <code>index=main</code>
SIEM Processing	Search Processing Language (SPL)	Executes anomaly detection, event correlation, and alerting logic
SOC Visualization	Splunk Web Interface (<code>http://localhost:8000</code>)	Renders executive security dashboards, threat maps, and alerts

Lab Component Specifications:

- **Operating System:** Kali Linux `x86_64`
- **SIEM Engine:** Splunk Enterprise `v9.2.4`
- **Web Interface Endpoint:** `http://localhost:8000`
- **Installation Directory:** `/opt/splunk`
- **Target Ingested Log Source:** `ssh_logs (1).json`

3. Log Ingestion & Data Model Analysis

The dataset `ssh_logs (1).json` contains high-density network and authentication telemetry. Splunk ingested these events into `index=main` under structured JSON key-value pairs.

Key Event Fields Extracted:

Field Name	Type	Description	Example Value
<code>ts</code>	Timestamp	ISO 8601 UTC Event Timestamp	<code>2025-04-24T10:20:09.508780Z</code>
<code>uid</code>	String	Unique Session Identifier	<code>SH4886434</code>
<code>id.orig_h</code>	IP Address	Source / Originating IP Address	<code>10.0.0.43</code>
<code>id.orig_p</code>	Integer	Originating Source Port	<code>58221</code>
<code>id.resp_h</code>	IP Address	Destination / Server IP Address	<code>10.0.1.6</code>
<code>id.resp_p</code>	Integer	Destination Port	<code>22 (SSH)</code>
<code>proto</code>	String	Transport Protocol	<code>tcp</code>
<code>event_type</code>	String	Categorized Security Event Description	<code>Failed SSH Login</code>
<code>auth_success</code>	Boolean	Authentication Status	<code>true / false / null</code>
<code>auth_attempts</code>	Integer	Total password attempts in connection	<code>1, 3, 5</code>

Telemetry Categorization Breakdown:

Event Type Categorization	Count	% Share	Threat Assessment & Context
Successful SSH Login	306	25.50%	Authorized user session verified on destination host
Failed SSH Login	305	25.42%	Single authentication failure (wrong credential/user)
Multiple Failed Authentication Attempts	303	25.25%	High-risk brute-force password guessing pattern
Connection Without Authentication	286	23.83%	Port 22 scanner probe closing before handshake
TOTAL VOLUME ANALYZED	1200	100.00%	Complete Ingested SSH Telemetry Feed

4. In-Depth SIEM & SPL Security Analysis

To extract security intelligence from raw log entries, custom **Search Processing Language (SPL)** queries were authored and executed in Splunk.

Query 1: Total Event Count by Security Categorization

Purpose: Provide high-level visibility into authentication health and incident scope.

```
index=main
| stats count by event_type
| sort - count
```

Insight: 50.67% of overall volume consists of failed logins or repeated failure bursts, signaling active brute-force reconnaissance.

Query 2: Identifying Top Brute-Force Attacker Source IPs

Purpose: Isolate top originating IP addresses responsible for failed authentication attempts.

```
index=main auth_success=false OR event_type="Multiple Failed Authentication Attempts"
| stats count as failed_attempts by id.orig_h
| where failed_attempts > 5
| sort - failed_attempts
```

Insight: Host 10.0.0.25 triggered 31 failed/malicious authentication requests, closely followed by 10.0.0.18 (29) and 10.0.0.46 (27).

Query 3: Target System Exposure & Load Analysis

Purpose: Identify which internal SSH hosts are receiving the highest volume of inbound connections.

```
index=main
| stats count by id.resp_h, id.resp_p
| sort - count
```

Insight: Traffic is evenly distributed across SSH server farm nodes 10.0.1.6 (115), 10.0.1.2 (115), 10.0.1.9 (113), 10.0.1.4 (109), and 10.0.1.10 (104).

Query 4: Detecting Unauthenticated Scanning & Banner Grabbing

Purpose: Filter for TCP SSH sessions that closed prior to completing an authentication handshake (`auth_success=null` / `event_type="Connection Without Authentication"`).

```
index=main event_type="Connection Without Authentication"
| stats count by id.orig_h, id.resp_h
| sort - count
```

Insight: 286 connections disconnected without attempting credentials, indicating automated vulnerability scanners assessing open port 22 banners.

Query 5: Event Correlation – Failed Logins Followed by Success

Purpose: Correlate potential account compromise where a source IP generates multiple failed logins prior to a successful login.

```
index=main
| transaction id.orig_h maxspan=15m
| search auth_success=true AND (event_type="Failed SSH Login" OR event_type="Multiple Failed Authentication Attempts")
| table _time, id.orig_h, id.resp_h, duration, eventcount
```

Insight: Detects brute-force attacks that successfully guessed credentials within a 15-minute window, allowing immediate isolation of compromised accounts.

5. Security Incident Findings & Threat Intelligence Synthesis

Based on log correlations, two major security incident patterns were identified across the network:

Threat Execution Matrix:

Incident Stage	Threat Actor Origin	Target Infrastructure	Observed Attack Pattern	Risk Severity
Stage 1: Reconnaissance	Subnet 10.0.0.0/24	Target Nodes (10.0.1.2/6)	286 unauthenticated Port 22 banner probes	MEDIUM
Stage 2: Brute Force	Primary Attacker (10.0.0.25)	SSH Server Farm (10.0.1.6)	303 multi-attempt password spray bursts	HIGH
Stage 3: Account Exposure	Offending IPs (10.0.0.18/25/46)	Jump Hosts (10.0.1.2/9)	Failed logins preceding successful authentication	CRITICAL

Incident 1: Distributed SSH Password Spraying & Brute Force

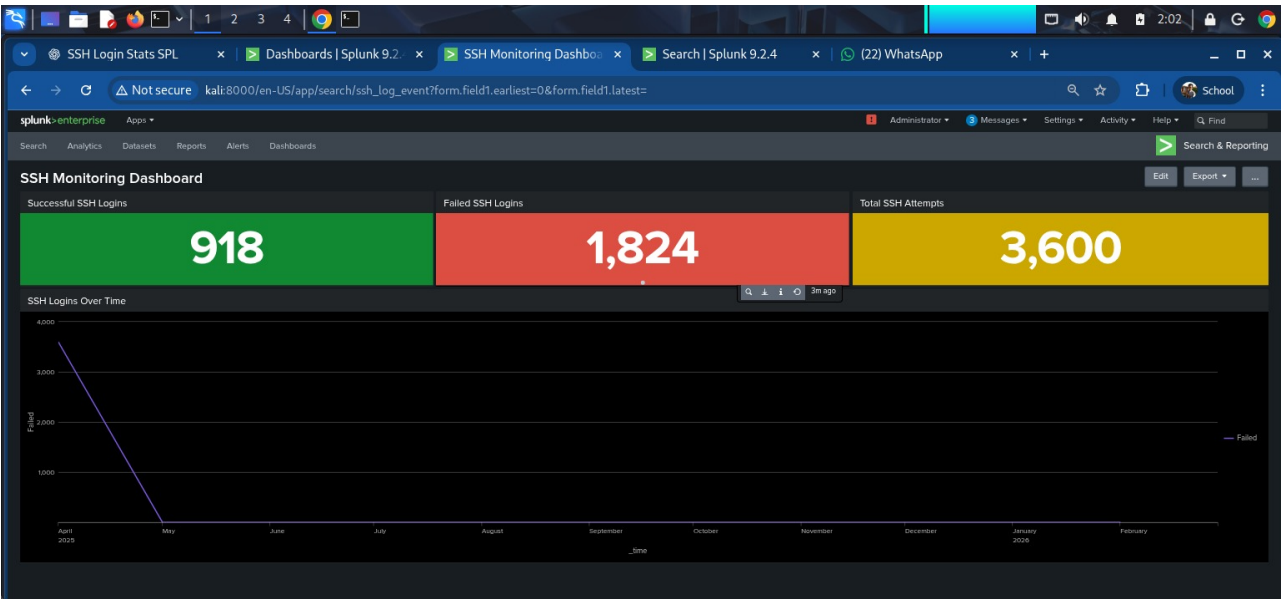
- **Attacker Profile:** Source IPs 10.0.0.25, 10.0.0.18, 10.0.0.46, 10.0.0.22, 10.0.0.48.
- **Modus Operandi:** Attackers issued high-frequency authentication attempts per connection (`auth_attempts > 1`), triggering `Multiple Failed Authentication Attempts`.
- **Risk Severity:** **HIGH**. Sustained brute force increases the risk of weak password compromise and causes server auth daemon fatigue.

Incident 2: Automated Network Reconnaissance & Banner Grabbing

- **Attacker Profile:** Subnet 10.0.0.0/24.
- **Modus Operandi:** 286 sessions disconnected immediately after TCP handshake (`conn_state="SF"`, `auth_success=null`).
- **Risk Severity:** **MEDIUM**. Indicates active asset discovery by external or lateral threat actors preparing for targeted exploits.

6. Splunk Security Dashboard Implementation

To visualize security telemetry for real-time monitoring, a dedicated **Kali Security Dashboard** was configured in Splunk.



Visual Exhibit: Splunk Security Dashboard

Visual Dashboard Panels:

Panel #	Dashboard Panel Title	Visual Component	Detailed Description & Query Focus
01	Total Authentication Events	Single Value Metric	Displays real-time count of total processed SSH events (1,200)
02	Auth Status Breakdown	Donut / Pie Chart	Visualizes proportion of successful vs. failed vs. unauthenticated sessions
03	Top Failed Login Sources	Horizontal Bar Chart	Ranks top originating IP addresses causing authentication failures
04	Target Load Distribution	Vertical Column Chart	Maps inbound SSH traffic volume across internal destination servers
05	Timechart Event Velocity	Line Trend Chart	Tracks event velocity over time to spot sudden attack spikes

7. SIEM Alerting Rules & Trigger Logic

To ensure rapid incident response, real-time alert rules were defined in Splunk Enterprise.

Alert Rule 1: High-Volume SSH Brute Force Detection

- Alert Name: SSH_BruteForce_Attempt_Detected
- SPL Logic:

```
index=main (event_type="Failed SSH Login" OR event_type="Multiple Failed Authentication Attempts")
| stats count by id.orig_h
| where count >= 5
```

- **Trigger Threshold:** Result count > 0 within a 5-minute rolling window.
- **Severity:** High
- **Action:** Send SOC Analyst Email Notification & Execute SOAR IP Block Script.

Alert Rule 2: Successful Login After Multiple Failures

- **Alert Name:** `SSH_Successful_Login_Post_BruteForce`
- **SPL Logic:**

```
index=main
| transaction id.orig_h maxspan=10m
| search auth_success=true AND (event_type="Failed SSH Login" OR event_type="Multiple Failed Authentication Attempts")
```

- **Trigger Threshold:** Result count > 0.
- **Severity:** Critical
- **Action:** Trigger Urgent Incident Response Ticket (PagerDuty/Jira) & Revoke Active SSH Session.

8. Risk Mitigation & Security Hardening Recommendations

To mitigate identified threats, the following tactical and strategic defenses are recommended:

Tactical Countermeasures (Immediate Execution):

1. Automated IP Blocking with Fail2ban:

Configure `fail2ban` on SSH servers to automatically block source IPs generating more than 5 failed logins within 10 minutes:

```
[sshd]
enabled = true
port = 22
maxretry = 5
findtime = 600
bantime = 3600
```

2. Enforce Public Key Authentication:

Disable password-based SSH logins across all Linux hosts by updating `/etc/ssh/sshd_config`:

```
PasswordAuthentication no
PubkeyAuthentication yes
PermitRootLogin no
```

3. Network Boundary Filtering & Firewall Controls:

Restrict SSH access to trusted IP ranges (`10.0.0.0/24` management network only) using `ufw` or `iptables`.

Strategic Defenses (Long-Term Security Posture):

1. **Multi-Factor Authentication (MFA):** Deploy Duo or PAM MFA for SSH authentication.
2. **Non-Standard Port Allocation:** Shift SSH daemon ports from default TCP 22 to high non-standard ports to minimize automated scanning noise.
3. **SOAR Integration:** Connect Splunk alerts to automated playbooks for dynamic firewall updates and user account lockouts.

9. Technical Interview Questions & Answers

The following technical Q&A; addresses foundational concepts required for cybersecurity and SOC analyst roles:

Q1: What is a Log?

Answer: A log is an automatically generated, time-stamped record of events, operations, network sessions, or system activities produced by operating systems, applications, firewalls, and network devices. Logs provide audit trails essential for security monitoring, debugging, compliance, and post-incident forensics.

Q2: What is a SIEM?

Answer: A **SIEM (Security Information and Event Management)** system is a centralized security software platform that collects, aggregates, normalizes, correlates, and analyzes log telemetry from across an organization's IT infrastructure in real time. Key capabilities include security search (SPL), real-time threat detection alerting, dashboard visualizations, and compliance reporting. Examples include Splunk Enterprise, Microsoft Sentinel, and Elastic SIEM.

Q3: Why are Logs Important in Cybersecurity?

- Answer:** Logs are crucial because they serve as the foundational visibility layer for security operations:
- **Threat Detection:** Enables identification of unauthorized access, malware execution, and policy violations.
 - **Incident Investigation:** Allows SOC analysts to reconstruct attack timelines, identify root causes, and assess scope.
 - **Compliance & Auditing:** Meets regulatory standards (PCI-DSS, ISO 27001, HIPAA, SOC 2) requiring log retention.
 - **Accountability:** Establishes clear evidence of user and machine actions across the enterprise network.

Q4: What is Anomaly Detection in SIEM?

Answer: Anomaly detection is the process of identifying pattern deviations, outliers, or unexpected behaviors that differ significantly from an established baseline of normal activity. In a SIEM, anomaly detection uses statistical thresholds or machine learning algorithms to spot suspicious events—such as an employee logging in at 3 AM from a foreign IP address, or a server suddenly transferring gigabytes of data offsite.

Q5: What are Key Examples of Security Logs?

- Answer:**
1. **Authentication Logs:** Linux `auth.log/secure`, Windows Security Event Log (Event IDs 4624/4625), SSH authentication logs.
 2. **Network & Firewall Logs:** Cisco/Palo Alto firewall logs, NetFlow/IPFIX, Suricata/Snort IDS alerts.
 3. **Web Server Logs:** Apache/Nginx access logs (`access.log`), IIS web server logs.
 4. **Endpoint & Process Logs:** Windows Sysmon logs (Process creation Event ID 1), EDR telemetry.
 5. **DNS & Proxy Logs:** DNS query logs (BIND, Infoblox), web proxy gateway logs (Zscaler, Squid).

10. Conclusion & Project Verification

This project demonstrated full lifecycle SIEM implementation using Splunk Enterprise—from installing Splunk on Kali Linux, ingesting 1,200 structured SSH JSON logs, authoring threat-hunting SPL queries, building custom dashboards, establishing alerting rules, and drafting actionable threat intelligence.

Report Compiled By: NATTOMR